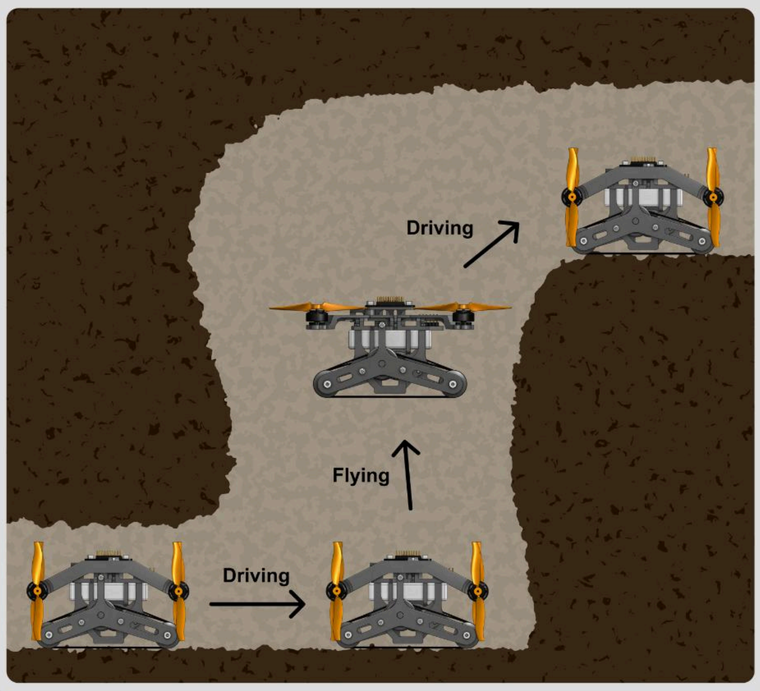
Palm-Scale Hybrid Aerial-Ground Robot for Confined and Unstructured Environments

Carnegie Mellon University Robotics Institute

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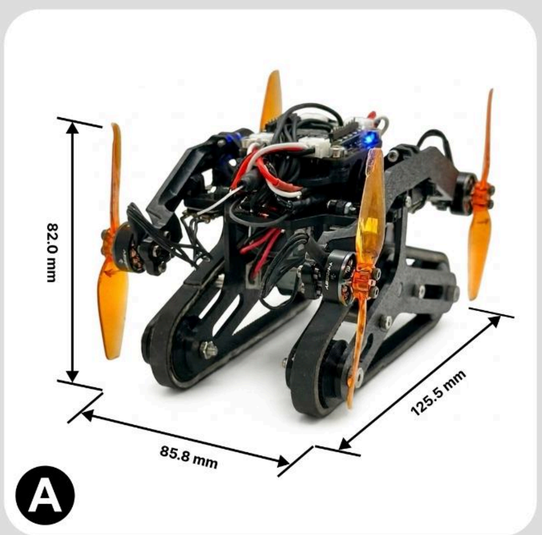
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Robot Overview



Rescuers need to see inside collapsed buildings and spaces too small or dangerous for people. Small ground robots get stuck at stairs and rubble. Small drones can fly over them, but their batteries run out fast. Our robot does both: it drives to save power and flies when something blocks its path. It fits in the palm of your hand, so it can reach places larger robots can't.

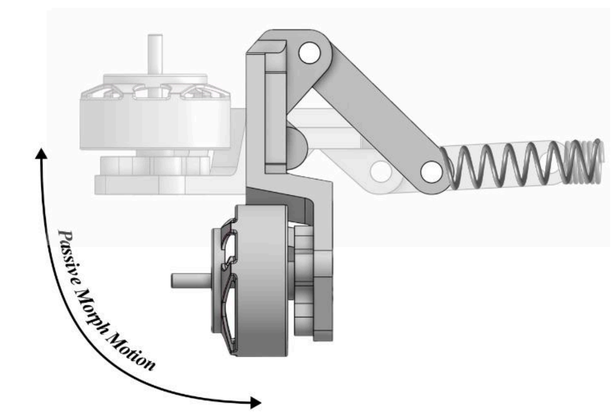
Design Overview



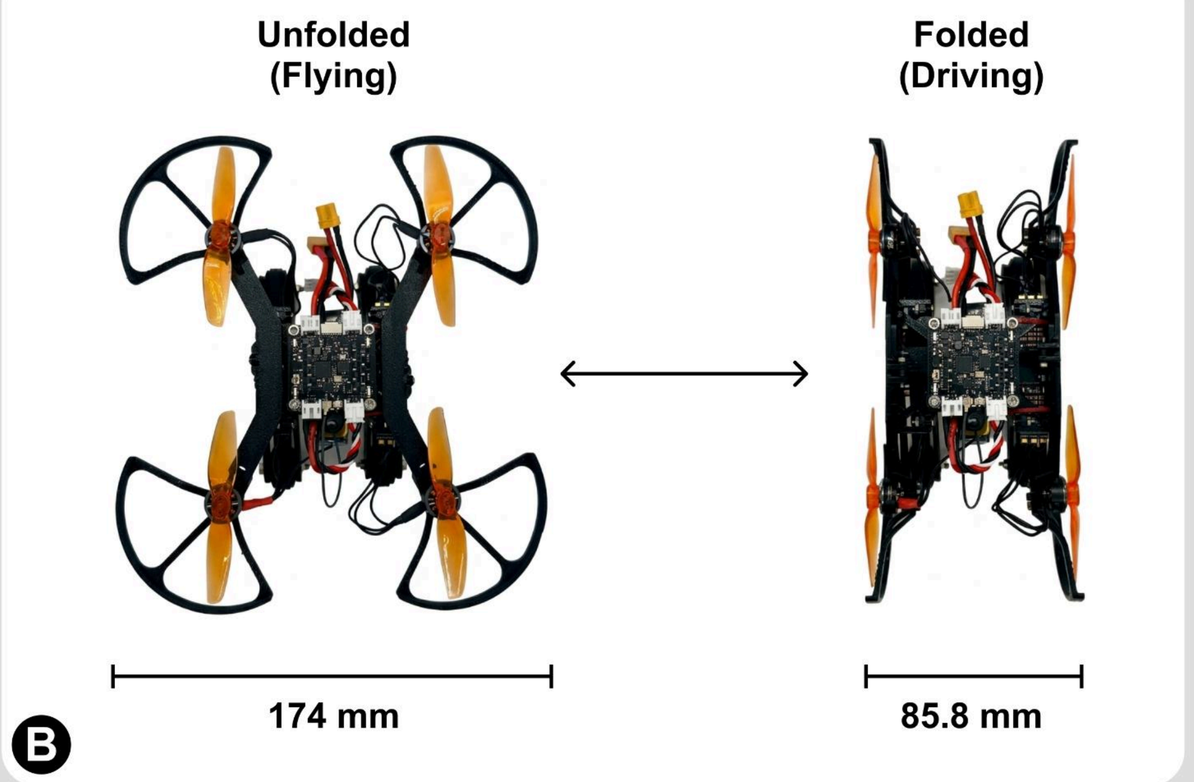


A palm-sized robot (85.8 × 125.5 × 82.0 mm) with four propellers for flight and two motor-driven tracks for driving. A spring-loaded linkage on each arm switches it between modes passively.

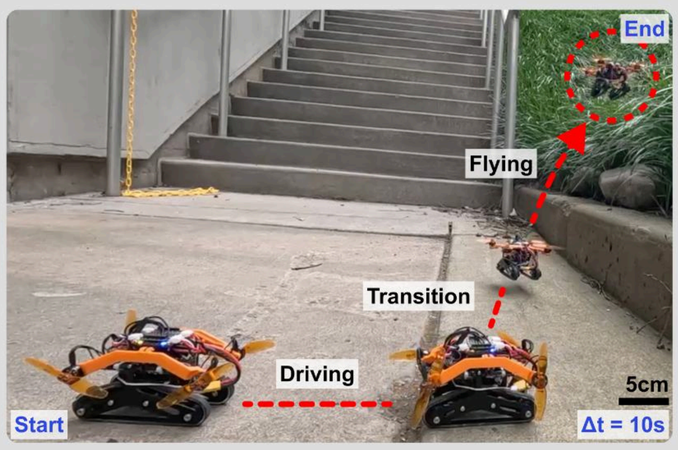
Bistable Mechanism



(A) shows the spring linkage on each arm. The spring holds the arm in one of two stable positions, open or folded, so no motor is needed. The arrow shows the path the motor swings through. Propeller thrust snaps the arm open for flight, and the landing impact folds it back down. (B) shows the result: the robot goes from 174 mm wide when flying to 83 mm when driving, less than half its width.

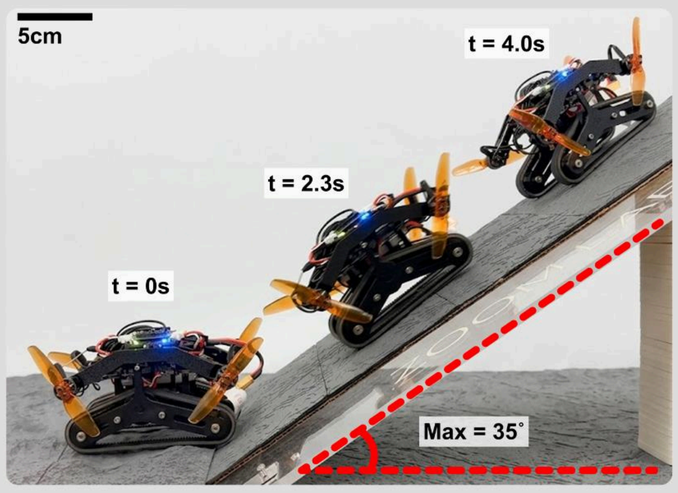


Multimodal Locomotion



The robot drives along the ground with its arms folded until it reaches a ledge it can not drive over. It spins up its propellers, which snap the arms open, then takes off and lands on the grass above. The whole run from start to finish takes 10 seconds.

Slope Traversal



With its arms folded, the robot uses only its tracks to climb a ramp. It can climb slopes up to 35° and reaches the top in about 4 seconds. Because it drives up instead of flying, it saves battery.

Contributions

- Palm-sized robot that drives and flies.
- Passive arms switch modes, no extra motors.
- Folds to under half its width: 174 → 83 mm.
- Climbs 35° slopes using only its tracks.
- Flies up ledges it can't drive over.
- Drives to save battery, flies when blocked.

Scan here to watch some videos of our drone!

We measured battery power in each mode. Driving on flat ground draws 8 W (1 A at 8 V). Hovering draws an estimated 48 W, about 6× more than driving. On one 450 mAh battery, the robot can drive for about 22 min but fly for only about 3.5 min.

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